

Gear Ratio Optimization via Quasi Steady-state Lap Time Simulation

Developed by S.Sarvesh

Project Overview-

This project is a custom-built Quasi-Steady State (QSS) Lap Time Simulator developed in MATLAB. It is designed to solve a multi-objective optimization problem for an Electric Vehicle (EV) race car: Finding the optimal gear ratio that perfectly balances minimum lap time with minimum energy consumption.

Unlike basic kinematic calculators (ideal case), this simulator treats the track as a continuous high-resolution micro-mesh ($dx = 0.1m$) and utilizes Bi-directional (Forward-Backward) Integration to autonomously calculate ideal braking points, apex speeds, and corner-exit accelerations.

Core Features-

- **3D Track Discretization:** Processes macro-level track data (Curve Radius, Arc Length, Elevation Gradients) into a continuous 70,000+ point mesh (for the track Spa Francorchamps taken as an example)
- **The "Anchor Point" Method:** Autonomously identifies the slowest point on the track (maximum lateral grip) to create a mathematically perfectly defined boundary condition for closed-loop track integration.
- **Dynamic Weight Transfer (Squat & Dive):** Calculates longitudinal load transfer based on the previous millisecond's acceleration, actively modifying front and rear tire normal forces (F_z).
- **2D Friction Ellipse (Circle of Traction):** Calculates the exact longitudinal grip available for acceleration and braking by subtracting the required lateral cornering force using the Pythagorean theorem.
- **EV Powertrain & Energy Model:** Integrates mechanical power over time, applying efficiency maps for Traction (draining battery).
- **Multi-Objective Optimization:** Uses Min-Max Normalization to generate a weighted cost function, plotting a Pareto Frontier to balance Lap Time (Performance) vs. Energy in kWh consumed (Endurance).

File Architecture-

Gear_optimization.mlx (The "Master" script) -

- Defines the macro-level track geometry (e.g., Spa-Francorchamps).
- Unrolls the track into the 0.1m step arrays for position, radius, and track gradient at that point.

- Calculates the absolute Aerodynamic Limit and the Global Grip Ceiling.
- Mathematically shifts the arrays to align with the Anchor Point.
- Executes a loop to sweep through gear ratios (e.g., 3.0 to 6.0), to find the lap times and energy consumed for each case.
- Normalizes the outputs, calculates the Cost Function, and plots the Pareto Frontier, to analyse the case of minimum lap time and the case of minimum cost such that there is a minimum possible lap time and energy consumed, with an optimal balance.
- The importance to optimise lap times and energy consumed can be given by assigning appropriate weights to the normalized values in the cost function to get desired results.

simulate_lap.mlx (The Lap Time Simulator function)-

- **Inputs:** current gear ratio, the unrolled position, radius and gradient arrays of the track, array of maximum possible speeds at calculated positions.
- **Forward Pass:** Integrates forward in distance. Calculates EV motor torque via interp1 RPM lookups. Determines acceleration by taking the minimum of the Power Limit (Motor) and the Traction Limit (Friction Ellipse + Weight Transfer).
- **Backward Pass:** Integrates backward in time. Calculates maximum braking deceleration based on aerodynamic drag, track gradient, and 4-wheel braking limits.
- **Merge:** Calculates the true velocity profile by taking the mathematical minimum of the Forward and Backward passes ($\min(v_{fwd}, v_{bwd})$). Calculates total lap time by finding time consumed in each of the small 0.1m segments and energy in kWh consumed.
- **Outputs:** Lap Time, Energy consumed in kWh.

A few results obtained-

For simulating a generic car of Formula 4 standards over the track Spa-Francorchamps:

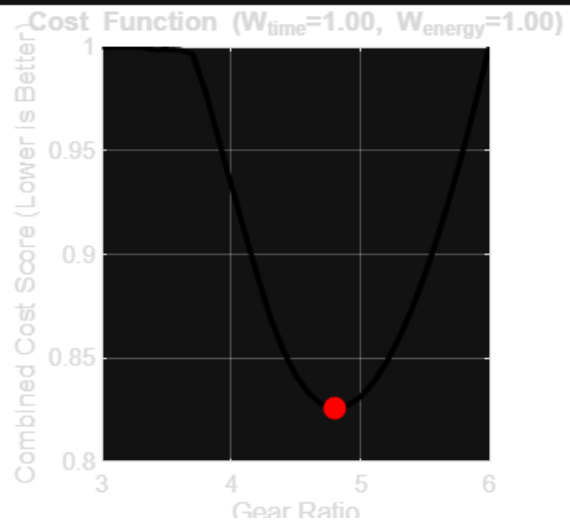
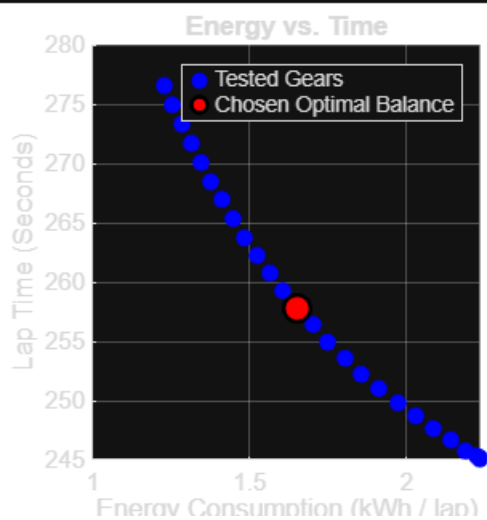
```
OPTIMAL GEAR RATIO: 3.10
FASTEST LAP TIME: 245.101 seconds
ENERGY USED: 2.233 kWh
=====
```



Starting Multi-Objective Gear Ratio Sweep

BALANCED OPTIMAL GEAR: 4.80

Lap Time: 257.786s | Energy: 1.653 kWh



Future scope and Improvements to be done-

1. Non-Linear Tire Modeling (Pacejka Magic Formula)

- **Current State:** Uses a static friction coefficient (μ).
- **Upgrade:** Implement the Pacejka '94 or '02 tire model to account for Tire Load Sensitivity. As aerodynamic downforce and weight transfer crush the tires, their efficiency (μ) drops. This will dramatically alter low-speed vs. high-speed grip limits.

2. 3D Aerodynamic Mapping

- **Current State:** C_d and C_{df} constant.
- **Upgrade:** Introduce a 2D lookup table that maps Drag and Downforce based on Front and Rear Ride Heights. As the car dives under braking, the aero balance shifts forward, changing the braking limits dynamically.

3. Lateral Load Transfer & Asymmetric Grip

- **Current State:** The car is modelled as a bicycle (longitudinal weight transfer only).
- **Upgrade:** Calculate lateral weight transfer during cornering based on Track Width and CG Height. Because tires are load-sensitive, the outside tires gaining weight cannot make up for the inside tires losing weight, lowering the overall v_{max} in high-G corners.